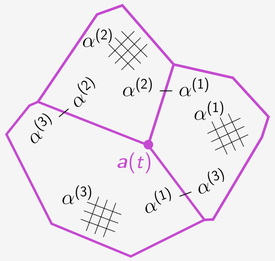
JKO-Based Framework for a Nonlinear Fokker-Planck Type Model of Grain Growth

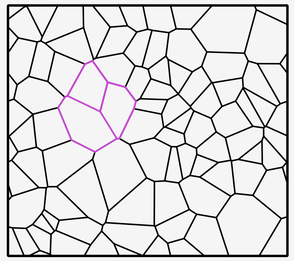
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What is Grain Growth?

- Grain growth is a highly complex multiscale-multiphysics process appearing in materials science which describes the evolution of the microstructure of polycrystalline materials [1, 8].
- These materials consist of many small monocrystalline grains which are separated by grain boundaries.
- The changes in grain size affect the material's electrical, thermal, etc. properties, which are important in the engineering of new materials.

Grain Boundary Network & Free Energy





- $a(t)$ is the triple junction point, $\alpha^{(j)} = \alpha^{(j)}(t)$'s are lattice orientations, and $\alpha^{(i)} - \alpha^{(j)}$'s are lattice misorientations.
- There is a coupled ODE system in terms of $a(t)$ and $\alpha^{(j)}(t)$, which we call the *vertex model* [6]. Since $N \gtrsim 10^4$ in applications, it is convenient to study $N \to \infty$ limit of the vertex model resulting in the following free energy with inhomogeneous absolute temperature $D(x)$:

$$F[f] = \int_\Omega \left(D(x)f(x,t)(\log f(x,t) - 1) + \phi(x)f(x,t)\right) dx$$

together with the dissipation relation

$$\frac{d}{dt}F[f] = -\int_\Omega \frac{f(x,t)}{\pi(x)} \left|\nabla\left(D(x)\log f(x,t) + \phi(x)\right)\right|^2 dx.$$

- The entropy-type free energy $F[f]$ is not λ-displacement convex for any $\lambda \in \mathbb{R}$, even for a convex or concave $D(x)$.

Nonlinear PDE Model

$$(\mathrm{nFP}) \quad \begin{cases} \dfrac{\partial f}{\partial t} = \nabla \cdot \left(\dfrac{f}{\pi(x)} \nabla \left(D(x) \log f + \phi(x) \right) \right), & x \in \Omega,\ t > 0, \\ \dfrac{f}{\pi(x)} \nabla \left(D(x) \log f + \phi(x) \right) \cdot \nu = 0, & x \in \partial\Omega, \\ f(x,0) = f_0(x), & x \in \Omega. \end{cases}$$

- $f(x,t)$ is probability density function,
- $\pi(x) > 0$ is mobility function,
- $D(x) > 0$ is absolute temperature function,
- $\phi(x)$ is energy density of a grain boundary.

JKO Scheme

We introduce the following Wasserstein quasi distance

$$c_\pi(x,y) := \pi(y)|x-y|^2,$$
$$W_\pi^2(\mu_1,\mu_2) := \inf_{\gamma \in \Gamma(\mu_1,\mu_2)} \left\{ \int_{\Omega\times\Omega} \pi(y)|x-y|^2 \, d\gamma \right\}.$$

Let $\tau > 0$ be a given time step size, and $f_0^\tau := f_0$. Then, for a given f_k^τ we recursively define f_{k+1}^τ as the minimizer of

$$(\mathrm{JKO}_{\mathrm{nFP}}) \qquad f_{k+1}^\tau \in \operatorname*{argmin}_{f \in \mathcal{P}(\Omega)} \left\{ \frac{W_\pi^2(f, f_k^\tau)}{2\tau} + F[f] \right\}.$$

This is a Jordan-Kinderlehrer-Otto (JKO) scheme [7, 2] assoociated with our free energy and dissipation relation.
We make the following assumptions:

- $\Omega \subset \mathbb{R}^d$ is bounded, connected, and C^1 domain.
- $f_0 \geq 0$, and $\int_\Omega f_0 dx = 1$.
- $F[f_0] < \infty$.
- $D \in C^2(\Omega)$ and $D \geq C_D > 0$.
- $\pi \in C^1(\Omega)$ and $0 < C_\pi^{\mathrm{low}} \leq \pi \leq C_\pi^{\mathrm{up}}$.
- $\phi \in C^1(\Omega)$.

At each step $k \geq 1$, there exists a unique minimizer of $(\mathrm{JKO}_{\mathrm{nFP}})$.

Euler-Lagrange Equation

For every tangential vector field ξ,

$$\frac{1}{\tau}\int_{\Omega\times\Omega} \pi(y)\xi(x)\cdot(x-y)\, d\gamma$$
$$= \int_\Omega Df_k^\tau \nabla\cdot\xi dx - \int_\Omega f_k^\tau(\log f_k^\tau - 1)\nabla D\cdot\xi dx - \int_\Omega f_k^\tau \nabla\phi\cdot\xi dx.$$

Moreover, for a Kantorovich potential φ_k,

$$D(x)\log f_k^\tau + \phi(x) + \frac{\varphi_k(x)}{\tau} = C_k \qquad \text{a.e.}$$

Idea of the Proof

For an optimal map T_k from f_{k-1}^τ to f_k^τ, we have

$$W_\pi^2(f_k^\tau, f_{k-1}^\tau) = \frac{\tau^2}{4}\int_\Omega \frac{f_k^\tau}{\pi(\mathrm{T}_k(x))} \left|\nabla\left(D(x)\log f_k^\tau + \phi(x)\right)\right|^2 dx$$
$$\geq \frac{\tau^2}{4C_\pi^{\mathrm{up}}} \mathrm{Diss}(f_k^\tau),$$

where

$$\mathrm{Diss}(f) = \int_\Omega (D(x))^2 \frac{|\nabla f|^2}{f} dx + \int_\Omega 2D(x)\nabla D(x)\log f\cdot\nabla f dx$$
$$+ \int_\Omega 2(\nabla D(x)\cdot\nabla\phi(x))f\log f dx + \int_\Omega 2D(x)\nabla\phi(x)\cdot\nabla f dx$$
$$+ \int_\Omega |\nabla\phi(x)|^2 f dx + \int_\Omega |\nabla D(x)|^2 f(\log f)^2 dx$$
$$=: I_1 + I_2 + I_3 + I_4 + I_5 + I_6.$$

Idea: Control the dissipation by the Fisher information. Since $I_5, I_6 \geq 0$, and $D(x) \geq C_D > 0$,

$$\mathrm{Diss}(f) \geq C_D^2\mathcal{I} + I_2 + I_3 + I_4, \qquad \mathcal{I} = \mathcal{I}[f] := \int_\Omega \frac{|\nabla f|^2}{f} dx.$$

It remains to prove $|I_2| + |I_3| + |I_4| \leq \frac{3}{4}C_D^2\mathcal{I} + C\left(\mathrm{Ent}(f) + 1\right)$. Then, this will imply Fisher-type spacetime regularity i.e., $\sqrt{f^\tau} \in L_t^2 H_x^1$.

Idea of the Proof (continued)

To estimate the integral I_2, we integrate by parts and get

$$I_2 = \int_\Omega \nabla(D(x)^2)\log f\cdot\nabla f dx$$
$$= -\int_\Omega f\log f\Delta(D(x)^2)dx - \int_\Omega \nabla f\cdot\nabla(D(x)^2)dx$$
$$+ \int_{\partial\Omega} f\log f\nabla(D(x)^2)\cdot\nu dS$$
$$=: I_{2,1} + I_{2,2} + I_{2,3}.$$

By Young's inequality,

$$|I_{2,1}| + |I_{2,2}| + |I_3| + |I_4| \leq \frac{C_D^2}{2}\mathcal{I} + C(\mathrm{Ent}(f) + 1).$$

The boundary integral $I_{2,3}$ requires a more careful estimation. First, we derive a trace inequality with ε. For any $u \in H^1(\Omega)$ and $\varepsilon > 0$,

$$\|u\|_{L^q(\partial\Omega)}^q \leq \varepsilon\|\nabla u\|_2^2\|u\|_2^{q-2} + C_\varepsilon\|u\|_2^q, \quad 2 \leq q < 2 + 2/d.$$

Second, we let $u := \sqrt{f}$, then for any $2 < q$, we have

$$|f\log f| = 2u^2|\log u| \leq C(q)u^q + 1.$$

Combining these two gives,

$$|I_{2,3}| \leq \frac{C_D^2}{4}\mathcal{I} + C.$$

Fisher-type Spacetime Regularity

For any $T < \infty$, the piecewise constant interpolation f^τ satisfies the following estimates

$$\left\| \sqrt{f^\tau} \right\|_{L^2(0,T;H^1(\Omega))}^2, \ \|f^\tau\|_{L^1(0,T;W^{1,1}(\Omega))} \leq C(1+T),$$
$$\|f^\tau\|_{L^{1+\frac{\alpha}{1+\alpha}}(\Omega\times[0,T])}^{1+\frac{\alpha}{1+\alpha}} \leq C(1+T).$$

Here, if $d > 2$, $\alpha := \frac{2}{d-2}$. If $d \leq 2$, $\alpha \in [1,\infty)$.

- The first estimate helps to apply Aubin-Lions-Simons type lemma from Dreher and Jüngel [4].
- The second estimate helps to get uniform integrability of $\{f^\tau \log f^\tau\}_{\tau>0}$ via de la Vallée Poussin's criterion.

Strong Compactness

For any $T < \infty$, and sequence of time steps $\{\tau_k\}_{k\geq 1}$ such that $\tau_k \to 0$ as $k \to \infty$, up to a (not relabeled) subsequence, we have

$$f^{\tau_k} \to f \text{ strongly in } L^1(\Omega\times[0,T]),$$
$$f^{\tau_k}\log f^{\tau_k} \to f\log f \text{ strongly in } L^1(\Omega\times[0,T]).$$

Weak Solution

Let f_0 be an initial data such that $F[f_0] < \infty$, and $\|f_0\|_{L^1(\Omega)} = 1$. The function $f \in L^1_{\mathrm{loc}}(\Omega\times[0,\infty))$ is a weak solution of (nFP) if

$$0 = -\psi(0)\int_\Omega \zeta(x)f_0(x)dx - \int_0^\infty\int_\Omega \partial_t\psi(t)\zeta(x)f(x,t)dxdt$$
$$- \int_0^\infty\int_\Omega \psi(t)\frac{1}{\pi(x)}\nabla\zeta(x)\cdot\nabla D(x)f(x,t)(1-\log f(x,t))dxdt$$
$$- \int_0^\infty\int_\Omega \psi(t)\nabla\cdot\left(\frac{\nabla\zeta(x)}{\pi(x)}\right)f(x,t)dxdt + \int_0^\infty\int_\Omega \psi(t)\frac{\nabla\zeta(x)}{\pi(x)}\cdot\nabla\phi(x)f(x,t)dxdt$$

holds for all $\psi \in C_c^\infty([0,\infty))$ and $\zeta \in H^1(\Omega)$.

Main Result

There exists a weak solution $f \in L^1_{\mathrm{loc}}(\Omega\times[0,\infty))$ of (nFP).

Discrete Maximum Principle

Assume that

$$f_0 \in \mathcal{P}^{\mathrm{ac}}(\Omega), \ \textit{and} \ \ D(x)\log f_0(x) + \phi(x) \in L^\infty(\Omega).$$

Then, any minimizer f_{k+1}^τ of $(\mathrm{JKO}_{\mathrm{nFP}})$ satisfies

$$\|D\log f_{k+1}^\tau + \phi\|_{L^\infty(\Omega)} \leq \|D\log f_0 + \phi\|_{L^\infty(\Omega)}.$$

The arguments of the proof are inspired by work of Tudorascu [9].

Exponential Convergence to Equilibrium

Recall from [5, Prop. 1.2] that the free energy minimizing equilibrium state of (nFP) is given by

$$f^{\mathrm{eq}}(x) := \exp\left(-\frac{\phi(x) - C^{\mathrm{eq}}}{D(x)}\right),$$

where C^{eq} is determined by the normalization. Let f be a smooth global in time solution of (nFP) with the initial data f_0 such that

$$D(x)\log f_0(x) + \phi(x) \in L^\infty(\Omega).$$

Then, we have

$$F[f] - F[f^{\mathrm{eq}}] \leq F[f_0]e^{-Ct}$$

Here, C depends on d, Ω, f_0, π, D, and ϕ.

Ongoing Work

In [3], we fully discretize $(\mathrm{JKO}_{\mathrm{nFP}})$ and obtain a structure-preserving numerical scheme. We also establish the convergence of the proposed numerical scheme.

Acknowledgments & References

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